Features

NetApp Solutions

NetApp
July 11, 2021

This PDF was generated from https://docs.netapp.com/us-en/netapp-solutions/containers/rh-os-n_use_case_advanced_cluster_management_features_cluster_lcm.html on July 11, 2021. Always check docs.netapp.com for the latest.

Table of Contents

Features 1
Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp 1
Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp 3
Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp 5
Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp 7
Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp 9

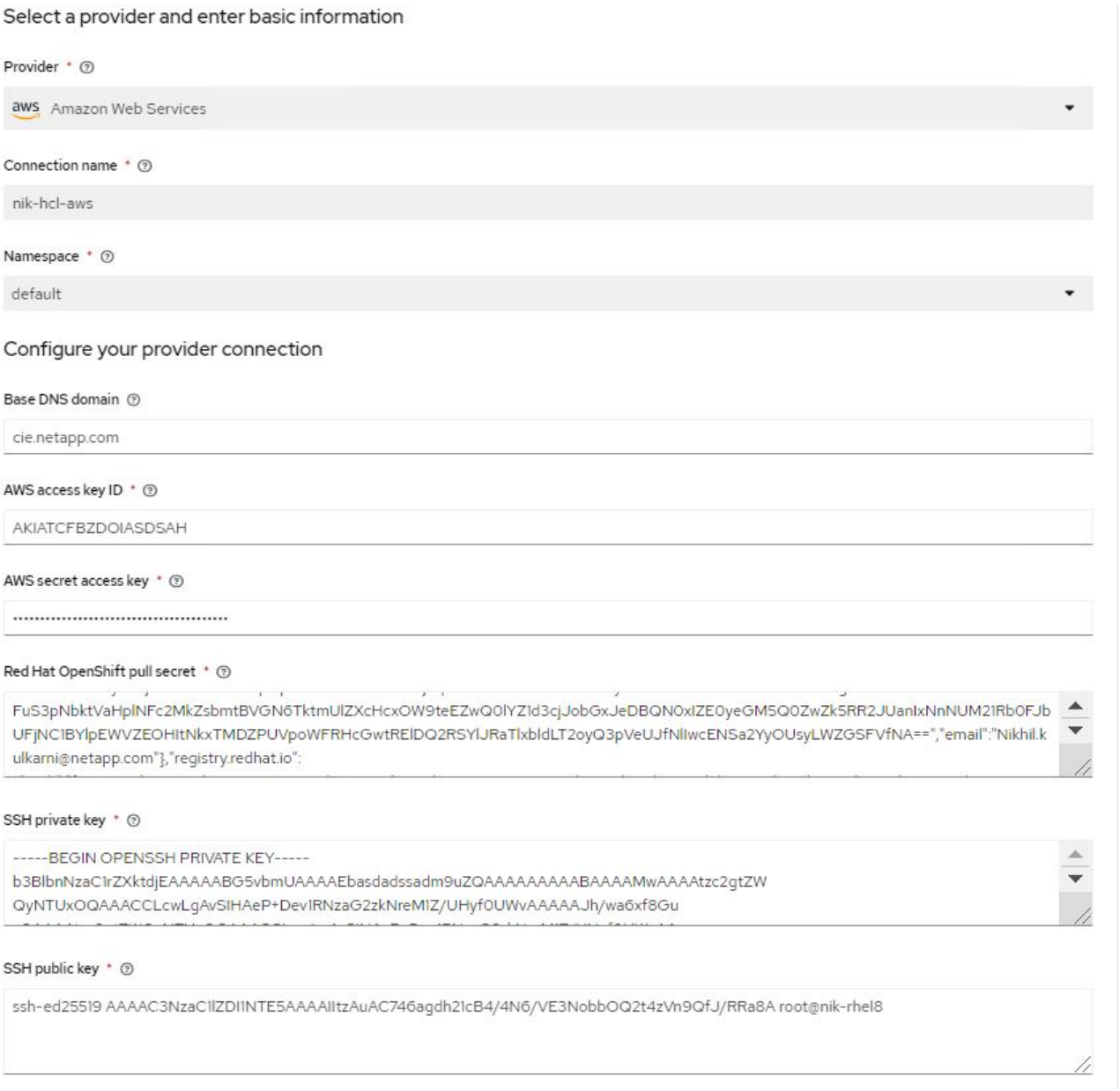
Features

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

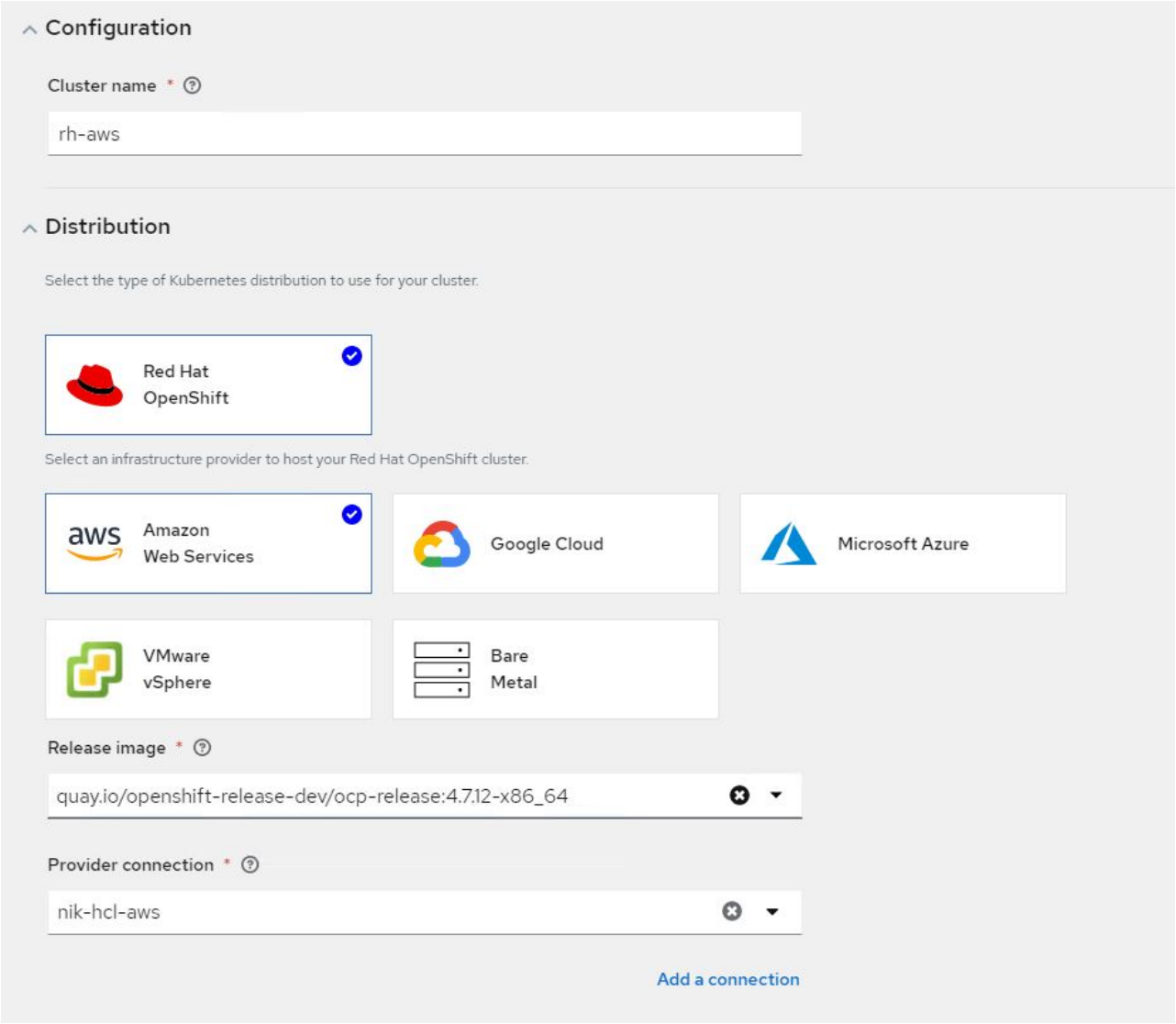
Cluster Lifecycle Management

To manage different OpenShift clusters, you can either create or import them into Advanced Cluster Management.

1. First navigate to `Automate Infrastructures` > `Clusters`.
2. To create a new OpenShift cluster, complete the following steps:
   a. Create a provider connection: Navigate to `Provider Connections` and click on `Add a connection`, provide all the details corresponding to the selected provider type and click on `Add`.



b. To create a new cluster, navigate to `Clusters` and click `Add a cluster` > `Create a cluster`. Provide the details for the cluster and the corresponding provider, and click `Create`.



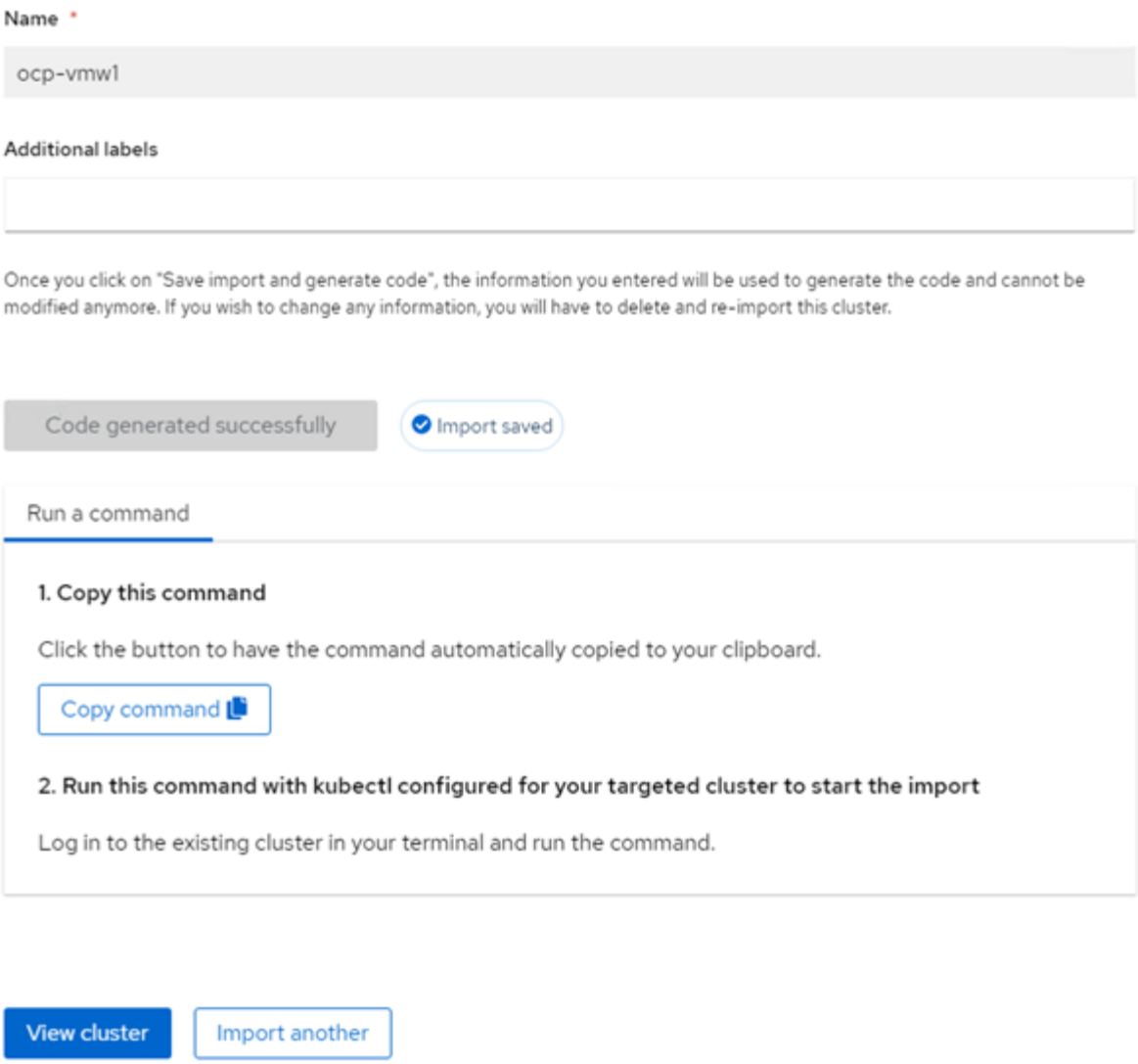
c. After the cluster is created, it appears in the cluster list with the status `Ready`.

3. To import an existing cluster, complete the following steps:

   a. Navigate to `Clusters` and click `Add a cluster` > `Import an existing cluster`.

   b. Enter the name of the cluster and click `Save import and generate code`. A command to add the existing cluster is displayed.

   c. Click `Copy command` and run the command on the cluster to be added to the hub cluster. This initiates the installation of the necessary agents on the cluster, and, after this process is complete, the cluster appears in the cluster list with status `Ready`.



4. After you create and import multiple clusters, you can monitor and manage them from a single console.

Next: Features - Application Lifecycle Management.

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

Application lifecycle management

To create an application and manage it across a set of clusters,

1. Navigate to `Manage Applications` from the sidebar and click `Create application`. Provide the details of the application you would like to create and click `Save`.

Applications /

Create an application YAML: Off

Cancel Save

Name*

demo-app

Namespace*

default

Repository location for resources

Repository types

Select the type of repository where resources that you want to deploy are located

Git

URL*

https://github.com/open-cluster-management/acm-hive-openshift-releases.git

Branch

main

Path

clusterImageSets/fast/4.7

2. After the application components are installed, the application appears in the list.

Applications

Refresh every 15s

Last update: 7:36:23 PM

Overview Advanced configuration

Create application

Search

Name	Namespace	Clusters	Resource	Time window	Created
demo-app	default	Local	Git		8 days ago

1 - 1 of 1 1 of 1

3. The application can now be monitored and managed from the console.

Next: Features - governance and risk.

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

Governance and Risk

This feature allows you to define the compliance policies for different clusters and make sure that the clusters adhere to it. You can configure the policies to either inform or remediate any deviations or violations of the rules.

1. Navigate to `Governance and Risk` from the sidebar.
2. To create compliance policies, click `Create Policy`, enter the details of the policy standards, and select the clusters that should adhere to this policy. If you want to automatically remediate the violations of this policy, select the checkbox `Enforce if supported` and click `Create`.

Governance and risk / Policies /

Create policy

YAML: Off

Name *

policy-complianceoperator

Namespace *

default

Specifications *

1 ✕ ComplianceOperator

Cluster selector

1 ✕ local-cluster: "true"

Standards

1 ✕ NIST-CSF

Categories

1 ✕ PR.IP Information Protection Processes and Procedures

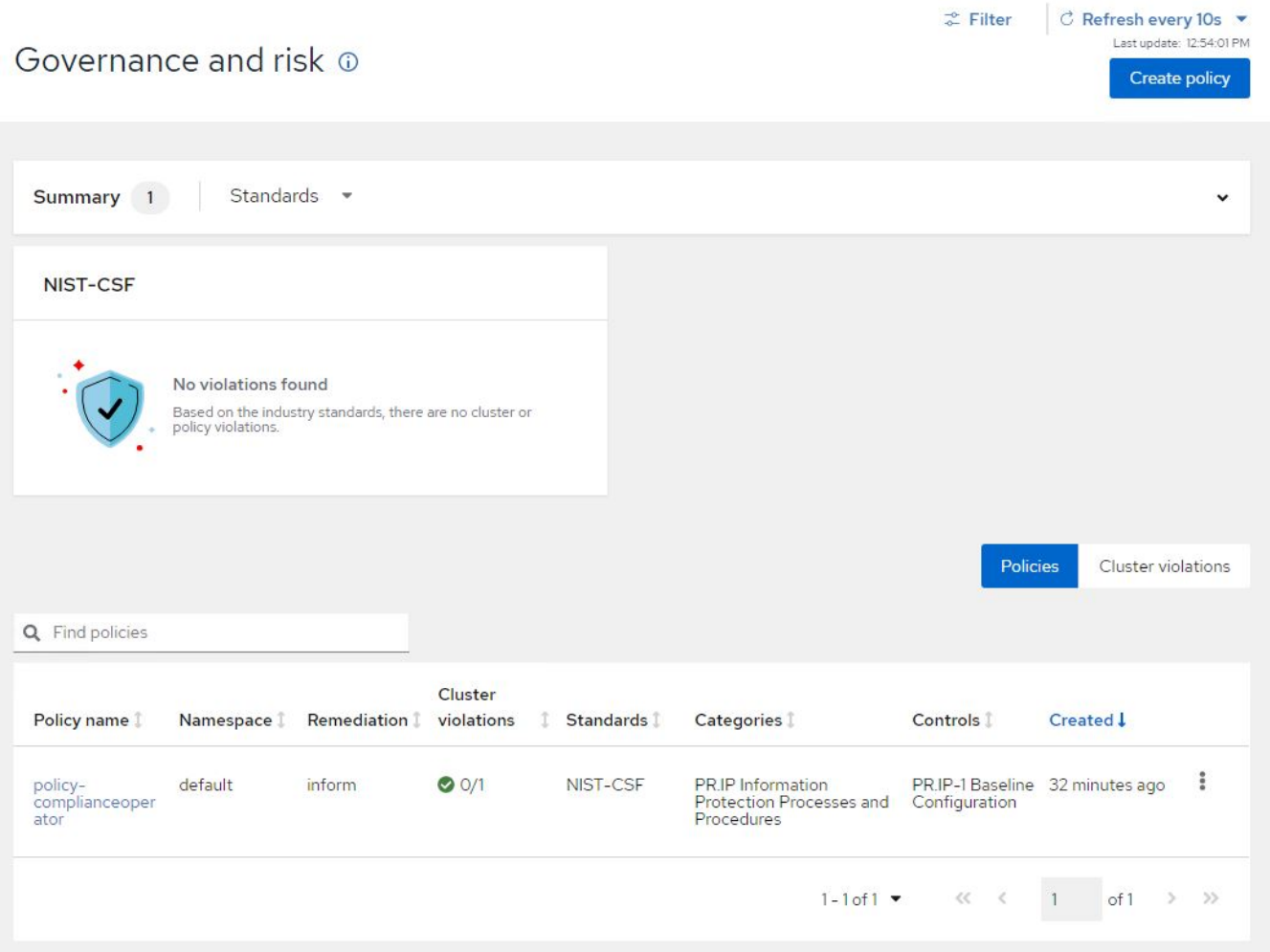
Controls

1 ✕ PR.IP-1 Baseline Configuration

☐ **Enforce if supported**

☐ **Disable policy**

3. After all the required policies are configured, any policy or cluster violations can be monitored and remediated from Advanced Cluster Management.



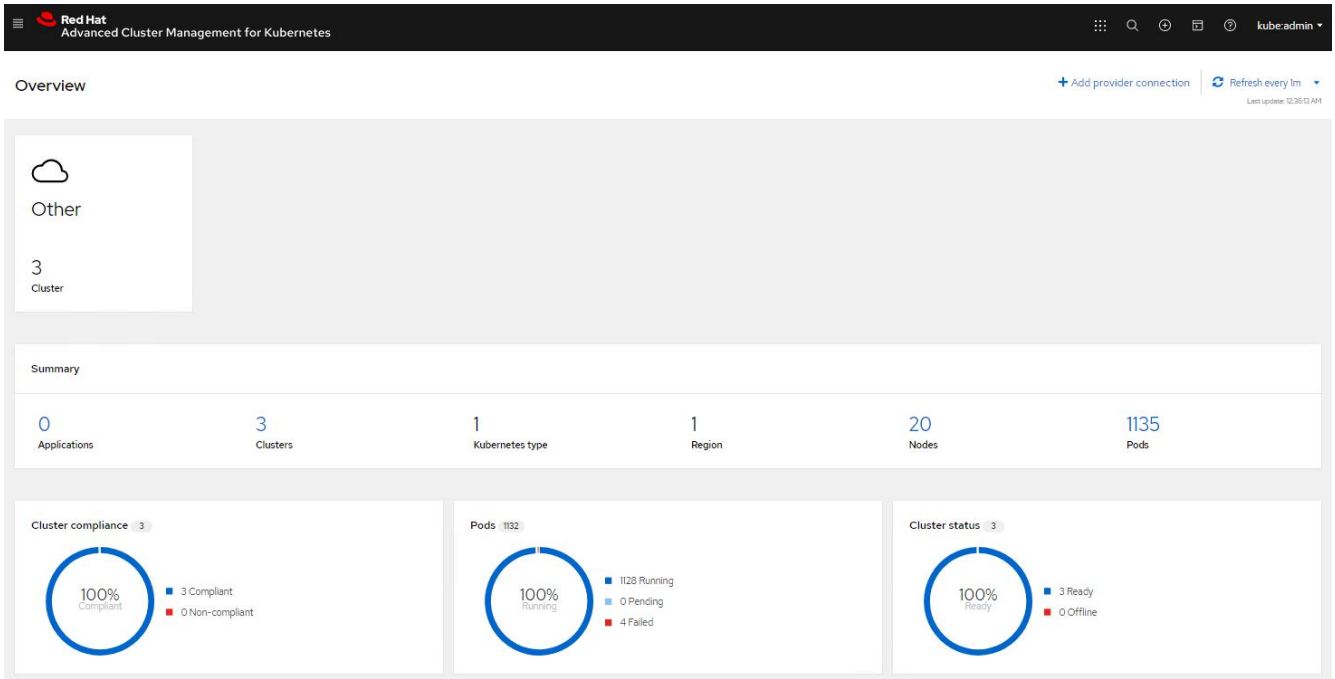
Next: Features - Observability.

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

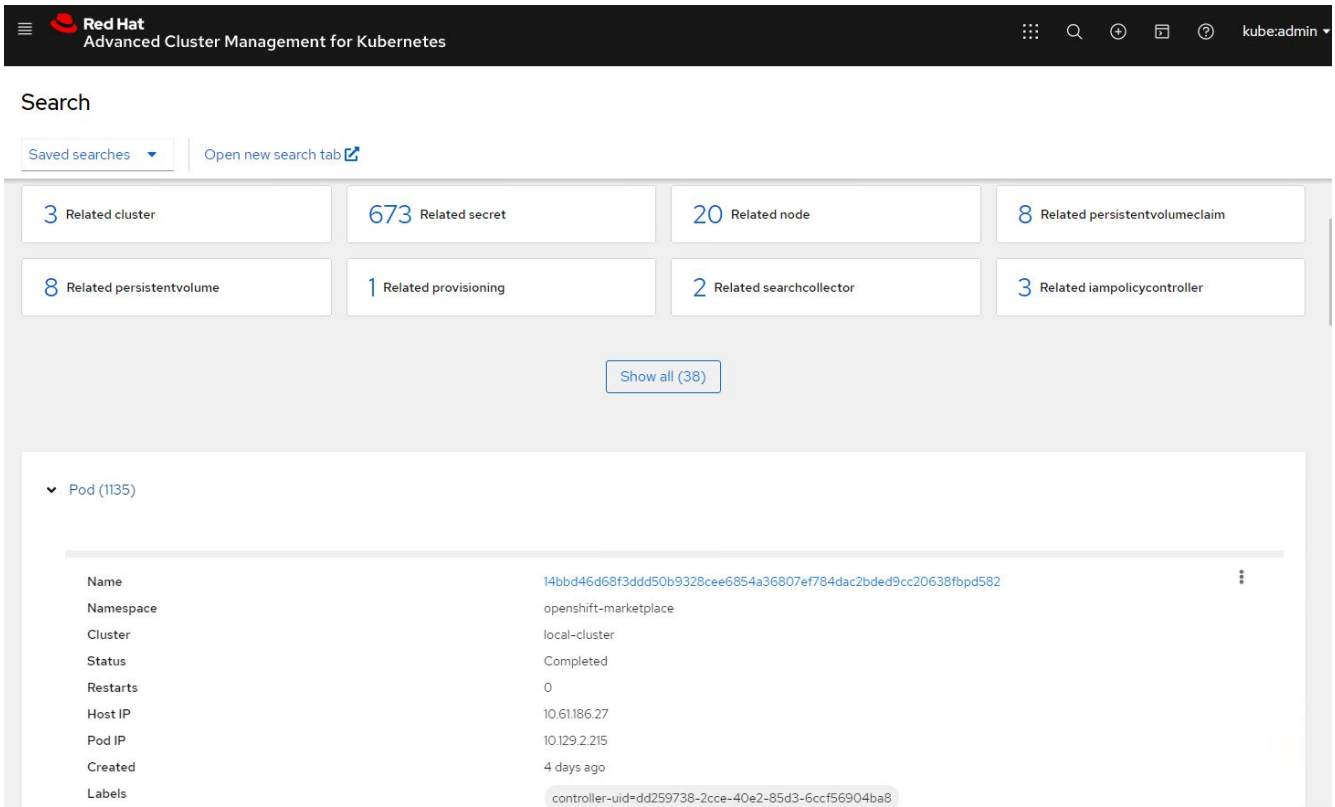
Observability

Advanced Cluster Management for Kubernetes provides a way to monitor the nodes, pods, and applications and workloads across all the clusters.

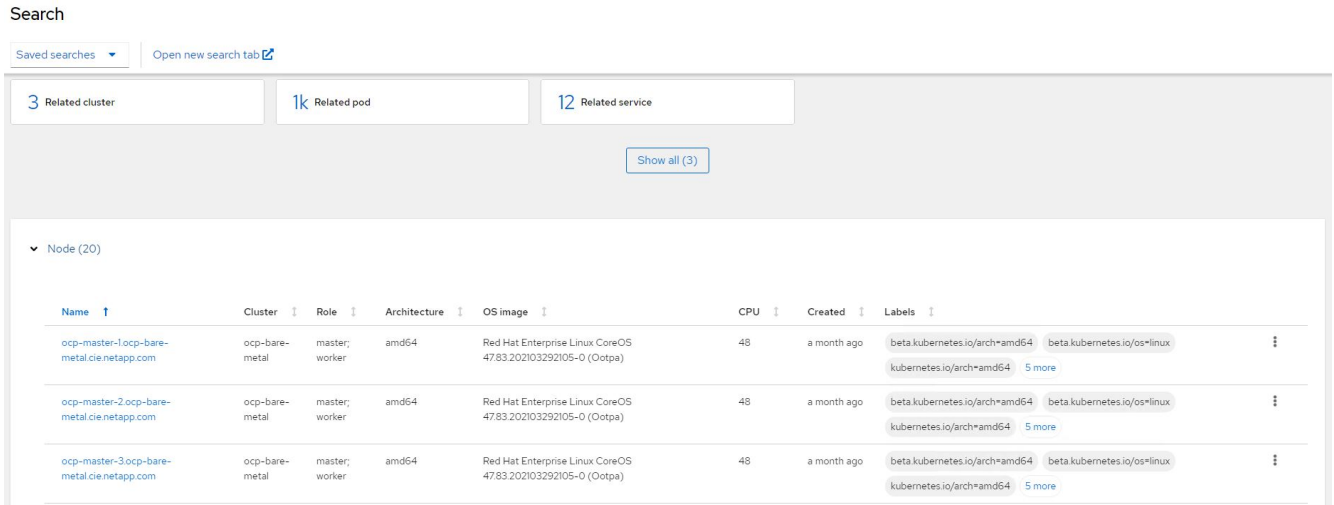
1. Navigate to `Observe Environments` > `Overview`.



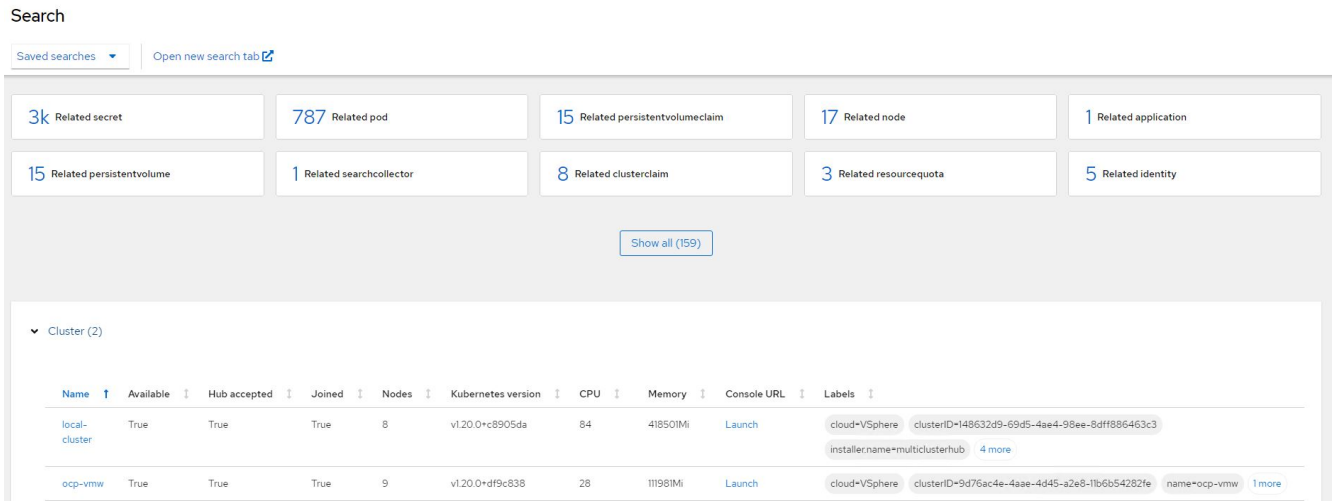
2. All pods and workloads across all clusters are monitored and sorted based on a variety of filters. Click `Pods` to view the corresponding data.



3. All nodes across the clusters are monitored and analyzed based on a variety of data points. Click `Nodes` to get more insight into the corresponding details.



4. All clusters are monitored and organized based on different cluster resources and parameters. Click `Clusters` to view cluster details.



Next: Features - Create Resources.

Features: Advanced Cluster Management for Kubernetes on Red Hat OpenShift with NetApp

Create resources on multiple clusters

Advanced Cluster Management for Kubernetes allows users to create resources on one or more managed clusters simultaneously from the console. As an example, if you have OpenShift clusters at different sites backed with different NetApp ONTAP clusters, and want to provision PVC's at both sites, you can click the + sign on the top bar. Then select the clusters on which you want to create the PVC, paste the resource YAML, and click `Create`.

Create resource

Cancel Create

Clusters | Select the clusters where the resource(s) will be deployed.

2 × local-cluster, ocp-vmw

Resource configuration | Enter the configuration manifest for the resource(s).

YAML

```yaml
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  name: demo-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
  storageClassName: ocp-trident
```

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.